Lesson 4: Mix It

Year 1/2 STEAM | Unit: Stronger Together: Paper Makers | CRT plan | 45 minutes

A little preparation (5 minutes). Check the class medical list for wheat or coeliac allergies. If there are any, use cornflour for the whole class. Cover each table with newspaper. Everything else is handed out by student helpers during the lesson. **No screens** are used.

This week each pair makes a **layered paper strip** that dries on the class tray for next week, and each student records a stick test in their **STEAM journal**. Journals are on the shelves as you walk in the door.

Learning Intention and Success Criteria

Learning Intention:

To find out that mixing flour and water makes paste, a new material with different properties.

Success Criteria:

I can:

- describe what flour and water are like on their own.
- mix them to make paste.
- test which one sticks paper best.

Read these aloud at the start and come back to them at the end.

What you need

Item	How many	Used for
Plain flour	1 bag (or cornflour)	A helper puts 2 heaped spoons into each pair's mixing cup
Water	1 jug	A helper pours a little into each pair's water cup
Cups	2 per pair	1 mixing cup, 1 water cup. Any small plastic or paper cups.
Spoons	2 per pair	Measuring and stirring
A4 scrap paper	2 sheets per pair, plus 1 per student	Test squares, strips and recording
Newspaper	A few sheets per table	Covering the tables
Paper towel	A roll	Wiping hands and tables
Glue sticks	1 per pair	Gluing the record sheet into journals
STEAM journals	1 per student	Shelves by the door

One flat tray or a spare table for the layered strips to dry overnight.

How to make paste and test it

1. Mix

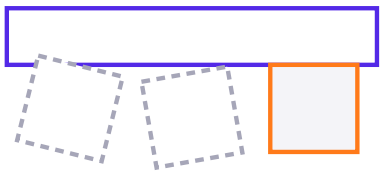
2 spoons flour, 2 spoons water



Stir 20 times. Count aloud.

2. Stick test

lift the strip and shake gently



flour water **paste**

Which squares stay on?

3. Layer it

strip, paste, strip, paste, strip



initials on the end in pencil

Onto the class tray to dry.

- **Describe first:** each student feels a pinch of flour between dry fingers (powdery, dry, soft) and looks at the water (runny, wet, it flows).
- **Make paste:** 2 spoons of flour into the mixing cup, then 2 spoons of water. Stir 20 times. Too thick? Add a few drops of water. Too runny? Add a pinch of flour.
- **Make test squares:** fold a sheet into quarters and tear on the folds to make 4 squares. Tear a long strip from the second sheet.
- **Stick test:** stick 3 squares onto the strip: one with a pinch of dry flour, one with a little water and one with paste. Press each for a count of 5. Lift the strip so the squares hang underneath and give it a gentle shake.
- **Layer it:** tear 3 more strips. Paste on the first, lay the second on top, paste again, lay the third on top. Write the pair's initials on the end in pencil and put it on the tray to dry.

Key words

powdery, runny, mix, stir, paste, sticky, new material, layer

Lesson sequence

Time	Step	What to do and say
0 to 3 min	Settle	Students sit on the floor facing you. Ask: *"Last week, how did we make paper stronger?"* (By weaving strips over and under.) Read the Learning Intention and Success Criteria aloud.
3 to 10 min	Hook	Hold up two small squares of paper. Ask: *"Could flour stick these together?"* Thumbs up or down. Sprinkle dry flour on one, press the other on top and lift: it falls off. Ask: *"What about water?"* Try again with a few drops of water: it slips off. Say: *"Flour can't do it. Water can't do it. Watch this."* Mix 2 spoons of flour and 2 spoons of water while the class counts the stirs to 20. Stick the squares with the paste and lift. It holds! Ask: *"What changed?"*

10 to 15 min	Describe and show how	Pass the flour bag around so each student feels a pinch. Write two lists on the board: flour is ___ (powdery, dry, soft, white), water is ___ (runny, wet, it flows). Write **mix** and **paste** on the board and say what they mean. Show the stick test and the layered strip once, saying each step aloud.
15 to 30 min	Make and test in pairs	Two helpers go table to table: one spoons 2 heaped spoons of flour into each pair's mixing cup, one pours a little water into each water cup. Other helpers hand out spoons and 2 sheets of paper per pair. Pairs make paste, do the stick test, then make one layered strip and put it on the tray. Roam with the questions below.
30 to 39 min	Clean up and record	Pack up first: spoons and cups to the sink, leftover paste scraped into the bin (not the sink), newspaper folded up, hands wiped. Give each student 1 sheet. They fold it in half and draw the 3 squares on the front, then tick the ones that stuck and cross the ones that fell. Helpers hand out journals. Students glue the sheet in and write the date and the title **Mix It**.
39 to 45 min	Share	3 or 4 pairs share using the sentence stem on the board. Finish with the key idea (below). Journals onto the shelves by the door.

Write this on the board

	Flour	Water	Paste
What is it like?	powdery	runny	sticky
Did it stick?	✓ or ✗	✓ or ✗	✓ or ✗

Sentence stem: **Paste is a new material because _____.**

Questions to ask while roaming

- What does the flour feel like? What does the water look like?
- What happened to the flour and water when you stirred them?
- Is paste like flour? Is it like water? How is it different?
- Which square stayed on? Why do you think it held?

Support and extension

Who	What to do
Support	Spoon the flour and water into the cup together, counting the stirs aloud. Students can record with ticks, crosses and drawings only.
Extension	Fast finishers (often Year 2) make a thick paste and a runny paste. Which one sticks better? Write one sentence explaining why.

Key idea to finish on: Flour on its own is powdery and water on its own is runny. Neither one sticks paper well. When we **mix** them, they make **paste**: a new material with new properties. It is sticky! Next week we will test the layered strips to see what the paste did to the paper.

Notes for the teacher

- **Allergies:** if anyone has a wheat or coeliac allergy, use cornflour for the whole class. It makes a paste that works the same way. If there is no cornflour, that student records the stick test for their partner and does not touch the flour.
- **Raw flour is not safe to eat.** Remind students not to taste the flour or the paste, and to keep their hands away from their mouths and eyes.
- A wet water square can cling for a moment. That is fine: it falls off once it dries, and the paste square stays stuck.
- Paste down the sink can block it. Scrape leftovers into the bin and rinse the cups.
- The layered strips need a flat spot to dry overnight. Please do not throw them away: next lesson starts by testing them.
- Please leave a short note about how far the class got and any students who were absent.

Curriculum links

- **Science VC2S2U05:** materials can be combined in a variety of ways for particular purposes; objects can have different properties from the materials they are made from.
- **Design and Technologies VC2TDE2C04:** explore the characteristics and properties of materials used to create designed solutions.
- **Maths:** measuring with informal units (spoons), counting to 20.